

Breaking Barriers: Providing Seamless Internet Across Different Providers

CS4991 Capstone Report, 2026

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ABSTRACT

Internet Service Providers (ISPs) only serve so far into Charlottesville, forcing some local companies that span multiple buildings to seek a variety of different ISPs for each location; creating an inconvenience for employees as they must reconnect to the networks from building to building. To resolve this issue, we must alter the settings of each provider's equipment to mimic the same network, allowing users to travel across buildings hassle-free. During my internship, we used UniFi equipment to set up network parameters and settings including SSID, security protocols, and general network information. We then installed the equipment in two separate buildings, one using Comcast and the other using Ting as their providers, completing the setup of the separate networks. By altering the settings, we successfully had the networks mirror one another, allowing users to travel from one building to another seamlessly without a gap in connection. Future work for these networks would include upkeep and possibly scaling up the software to include more than two buildings.

1. INTRODUCTION

The BridgeLine company in Pantops, VA, has a contract with Seventh Wall, a Cybersecurity and IT company I am currently interning with. In October 2025, The BridgeLine reached out about updating their networking equipment. The company is

composed of two buildings, the main building using Ting as their provider and the secondary building using Comcast.

The primary providers of Charlottesville and the surrounding area are both Ting and Comcast (Xfinity). Ting is more common for residential use due to its price point for high internet speeds, while Comcast offers affordable scalability and reliability and provides more desirable plans for businesses. Seventh Wall uses Ubiquiti's UniFi platform to manage and monitor client networks.

When The BridgeLine added their secondary building, they were unable to use the same network established in the main building since Ting does not service the area where the second building is located, despite the two buildings being across the street from one another. Their Wi-Fi signals even reach far enough to provide a faint signal across the two buildings. This makes clear connectivity in the buildings difficult, creating a daily problem as employees travel between the two sites frequently.

2. RELATED WORKS

While literature on the method we used to mirror the networks is not extensive, there is literature on the factors and considerations that led to our final solution. When establishing networks for multi-site companies, the most essential traits to look for from providers are

scalability, reliability, persistence and price [2, 3, 4]. When developing a solution for a company there is no one right answer, as every company has different needs [1]. Strickland [2025] suggests businesses should use one network provider, however, for The BridgeLine the suggestions of s2s Communications [2025] make a more compelling case.

s2s Communications' [2025] push for companies to use different providers across their sites for increased persistence and reliability. If one provider goes dark, a company can just move to their second site without losing internet connection [2]. In the case of The BridgeLine, they did not have a choice between using the same or different providers. Nonetheless, outages are common in Pantops, making two providers better than one for the same reasons outlined in s2s Communications [2025].

Not only does Seventh Wall already use UniFi services, Ubiquiti [n.d.] provides many case studies that support this choice of product in all cases, including The BridgeLine. One of these case studies concerns a large company that needed strong, seamless connection across 50 offices and roughly 400 unique users [3]. They decided to use Ubiquiti's UniFi equipment due to its reputation for reliability and scalability, all at a reasonable price [3]. These factors were detailed in both Totino [2025] and s2s Communications [2025] as imperative for multi-site companies. This value alignment with UniFi products reinforces the use of their equipment in the network upgrade for The BridgeLine.

3. PROJECT DESIGN

This project was split into three stages to fully deploy the system: ideation, offsite preparation, and onsite installation.

3.1 Ideation

Before beginning work on The BridgeLine's networks, we had to assess their existing system and see how we could optimize their networks, as well as address their concerns. The BridgeLine was already operating with separate providers on their two sites. This means that we needed, at a minimum, double the amount of networking equipment as a single-site and provider business. We had to determine what equipment was necessary for the project and confirm pricing, as well as communicate goals and expectations with The BridgeLine. After confirming implementation expectations with The BridgeLine, we ordered the necessary networking equipment.

3.2 Offsite Preparation

Prior to installation within The BridgeLine's buildings, we were able to set up their networking equipment and protocols, to allow for quicker installation onsite and ensure that the UniFi equipment delivered was not defective. The pre-configuration of these devices included both testing the equipment on our network and establishing the network policies within the UniFi dashboard. We established the Wide Area Network (WAN) interfaces as well as Local Area Network (LAN) IP addressing schemes of primary and guest networks for both sites, these values serve as baseline for IP addresses of connected devices. We needed to ensure separate networks for guests of The BridgeLine so unauthorized hosts could not access important devices that remain consistently connected to the primary network, such as printers. To do this, we created separate virtual LANs on the networks to separate them using network firewall rules.

In addition to LAN and WAN initialization, we needed to define the SSID for primary and guest networks at each site. The SSID was the glue to our solution, as it is what devices refer to when searching for networks and how each network is identified by hosts. We had two

options for the sites: keep the two networks uniquely separate through separate SSIDs or give them the same SSID to make them recognizable as the “same” network, even though they originate from different providers. We decided on the latter to eliminate the headache of unique network identifiers and assigned the networks the same SSIDs and filler passwords, until The BridgeLine decided new permanent ones.

3.3 Onsite Implementation

After all preparations were complete, we traveled onsite to both of The BridgeLine’s sites to install the networking equipment. We first hooked up the hardware on their main site serviced by Ting. Ideally, we would have set up the hardware on the main site and done a mostly plug-and-play implementation since we had already set up and tested the hardware prior to traveling onsite. However, Ting was experiencing an outage in their Pantops/Charlottesville region at the time that we attempted to connect our UniFi equipment to its network. A representative confirmed the unfortunately timed outage.

In view of the outage, we continued to the second site, where we fortunately experienced the plug-and-play implementation we had initially hoped for on both sites. Comcast was not experiencing any outages or connectivity issues at the time of the installation. This was essential for The BridgeLine’s operations, as they could not afford to fall behind on their duties due to a network outage. Aside from a difficult printer, the second site’s installation went much more smoothly than the main site. The only task left was to confirm the success of our implementation once Ting came back online. We did not need to remain onsite for this.

4. RESULTS

After Ting came back online and we ensured the networks were operating as expected, we

found that the matching SSID implementation worked exactly as we anticipated, providing ease for employees who frequent both buildings or only travel between the two occasionally. The BridgeLine was satisfied with their new networks and, with typical upkeep of their new setup, we anticipate it to be a solution that will remain for the foreseeable future.

In addition to the success of the SSID solution, this project also highlighted the importance of multi-site businesses in high outage areas. Because The BridgeLine has two providers, even if not by choice, they are able to continue normal operations in the event of a Ting or Comcast outage. For companies with only one provider, the length and severity of network outages can cost them time and money.

This project has been performing well since the installation of the new UniFi equipment. It has made day-to-day operations simpler for The BridgeLine and has given Seventh Wall more direct access to their equipment via the UniFi Dashboard we use to monitor network status.

5. CONCLUSION

The needs of The BridgeLine presented a system with higher complexity than a typical network installation. This provided Seventh Wall with the opportunity to move forward with a creative implementation of The BridgeLine’s network. In addition, it gave me the opportunity to participate in a project that I would not have otherwise experienced. The implementation process gave me hands on experience with networking policies and equipment, that I will take forward with me into a professional career. Additionally, this project has given Seventh Wall another case study to market their services and flexibility to complex situations. The success of this project will help drive further business both through recommendation from The BridgeLine, as

well as from internet traffic on Seventh Wall's website.

6. FUTURE WORK

Future work for this project includes two main possibilities. The first possible expansion of this project is to fully bridge the networks together. Right now, the networks mimic one another but are not talking to each other. The way in which we "connected" the networks allows for bridging via Virtual Private Network (VPN). To do this we would implement a VPN across both sites that acts as a bridge between the Ting and Comcast networks allowing users to remain connected continuously rather than moving from one to another. Though, this implementation would make more sense if both networks existed in the same building. The second possibility for expanding this project would be the addition of more buildings to The BridgeLine. This would consist of completing the same implementation steps as previously mentioned.

7. ACKNOWLEDGMENTS

Thank you to Chad Reimold and Chris Hewitt of the Seventh Wall team for fostering this project and paper.

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